

Explainable-by-Design Speech Emotion Recognition via Concept Bottleneck Models on Self-Supervised Speech Representations

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ABSTRACT

Speech emotion recognition (SER) built on self-supervised (SSL) speech representations is accurate but opaque. We present an explainable-by-design SER pipeline that inserts a concept bottleneck between a frozen SSL backbone (HuBERT or wav2vec 2.0) and the emotion classifier: 15 acoustic concepts in the spirit of eGeMAPS, measured directly from the signal, predicted by linear probes, and used as the only input of the final classifier. Rather than maximizing accuracy, we quantify what interpretability costs and what buys it back. On IEMOCAP (speaker-independent, four classes), the bottleneck variants form a hierarchy (hard, soft, continuous, Concept Embedding Model) that trades explainability for accuracy on the same representation, from ≈ 0.67 accuracy for the unconstrained baseline to ≈ 0.47 for the concept bottleneck. A double-oracle decomposition shows that the cost is dominated by concept-set incompleteness (≈ 20 accuracy points) rather than by binarization (≈ 5 points). Targeted concept interventions localize the information leakage of soft concept scores exactly in the probes' errors. Explanation quality is assessed with the Co-12 framework, and all key findings replicate across the two backbones.

KEYWORDS

Explainable AI, Concept Bottleneck Models, Speech Emotion Recognition, Self-Supervised Learning, HuBERT, wav2vec 2.0, Concept Interventions, IEMOCAP

1 INTRODUCTION

Speech emotion recognition is increasingly deployed in sensitive settings, such as call-center triage, health monitoring, and human-machine interaction, where accuracy alone is not enough: users and auditors need to know *why* an utterance was classified as it was, in terms they can inspect and contest.

Modern SER is dominated by self-supervised speech models such as wav2vec 2.0 [2] and HuBERT [6], whose representations achieve strong accuracy with simple downstream heads but offer no human-understandable account of their decisions. Explainability research for audio also lags behind vision and text: there are no standard concept vocabularies for paralinguistics, and explanations must be meaningful with respect to the temporal and spectral structure of the signal.

This work does not aim to maximize accuracy. Instead, we build a concept-based, explainable-by-design SER system and *measure* both the price of interpretability and what buys it back. Our contributions are:

- a CBM pipeline on frozen SSL representations for SER, whose concepts are acoustic quantities measured per-sample from the signal itself, rather than derived from concept banks or vision-language models;

- a quantitative decomposition of the interpretability cost into a binarization component (≈ 5 points) and a concept-set incompleteness component (≈ 20 points);
- a controllability analysis with standard and *targeted* concept interventions, which localizes the information leakage of soft concept scores in the probes' errors;
- a comparison of Concept Embedding Models against CBM variants and of two SSL backbones.

2 RELATED WORK

2.1 Speech Emotion Recognition with Self-Supervised Representations

wav2vec 2.0 [2] learns speech representations by contrastive prediction over quantized latents; HuBERT [6] by masked prediction of clustered hidden units. Both transfer to paralinguistic tasks with frozen features and light heads. IEMOCAP [4] is the standard SER benchmark; SUPERB [15] reports emotion-recognition accuracy around 0.65 for linear probing of frozen base models, which anchors our black-box baselines. These models are accurate but opaque: nothing in their embeddings is directly readable by a human.

2.2 Concept-based and By-Design Explainability

Concept-based XAI explains predictions through human-understandable attributes [13]. Concept Bottleneck Models [9] predict concepts first and the label only from them, in independent, sequential, or joint training regimes, and support test-time *interventions* that replace predicted concepts with ground truth. Concept Embedding Models [17] replace each scalar concept with a pair of active/inactive embeddings, closing the accuracy gap while retaining concept supervision and interventions; the channel, however, is wider and less constrained (Section 5). TCAV [8] represents concepts post-hoc as directions (CAVs) in a network's activation space; Post-hoc CBMs [16] build a concept bottleneck on top of a frozen backbone using concept banks or CLIP.

Our model is a *sequential CBM realized post-hoc* on a frozen SSL backbone, following PCBM: "post-hoc" refers to how concepts are read out of a pre-trained representation, "by-design" to the final classifier, which sees nothing but concepts. Unlike PCBM, our concept labels are *measured from the signal* for every sample (no concept bank or validated audio-language model exists for paralinguistics), so per-sample interventions are possible; and we use no residual connector, the accuracy-recovery role being played by the CEM. Technically our probes *are* CAVs; what differs is the provenance of the labels, measured rather than curated.

2.3 Evaluating Explanations

Explanations must be evaluated, not just displayed. We adopt the Co-12 framework [12] plus randomization sanity checks [1]; plausibility expectations come from the acoustic-profile literature on vocal emotion [3, 7, 14].

3 RESEARCH GAPS

First, there is no standard taxonomy of concepts for paralinguistic tasks: vision has curated concept banks, audio does not. We address this with 15 acoustic concepts measurable from the signal, turning parameter sets such as eGeMAPS [5] into an intervenable concept vocabulary. Second, XAI for audio is under-explored compared to vision and text, and rarely evaluated beyond anecdotal saliency; we run a full Co-12 evaluation, including negative results, on both backbones. Third, concept-based SER and controllability via interventions are almost unexplored; we provide an intervention analysis with a novel targeted policy that doubles as a diagnostic for concept leakage.

4 METHODOLOGY

4.1 Problem Formulation and Overview

Given an utterance x , a frozen SSL encoder produces layer-wise representations $h_1(x), \dots, h_{12}(x)$; a learned aggregation yields an embedding $z(x)$; probes map z to interpretable concepts $\hat{c}(x)$; a linear head maps concepts, and concepts only, to the emotion $y \in \{\text{neutral, happy, sad, angry}\}$ (Figure 1). Only the 12 aggregation weights, the probes and the heads are trained.

4.2 Concept Set

The bottleneck uses $K=15$ acoustic concepts in the spirit of eGeMAPS [5], grouped into prosody, energy, and voice quality (Table 1). All are computed per utterance from the trimmed waveform (librosa [11] and Praat), hence available for every sample at deployment.

Two further sets are kept strictly out of the main model: the first four mel-frequency cepstral coefficients (MFCCs), excluded because they are not human-interpretable and retained for ablation only, and IEMOCAP’s valence/activation/dominance annotations, which are quasi-labels of the target and serve only as a diagnostic upper bound.

4.3 Frozen SSL Backbone and Layer Aggregation

The backbone (HuBERT-base or wav2vec 2.0-base) is frozen. Utterance embeddings are mean-pooled per layer and combined as

$$z(x) = \sum_{l=1}^{12} \alpha_l \text{LN}_l(h_l(x)), \quad \alpha = \text{softmax}(a), \quad (1)$$

where the per-layer LayerNorm is applied *before* the weighted sum; without it, deeper layers dominate through their larger norms. Only the 12 weights a are trained (cross-entropy, early stopping on the internal validation session).

4.4 Concept Bottleneck and Classifier

Each concept has a logistic probe on z ; the regularization strength is selected per concept on the validation session and the probe is refit

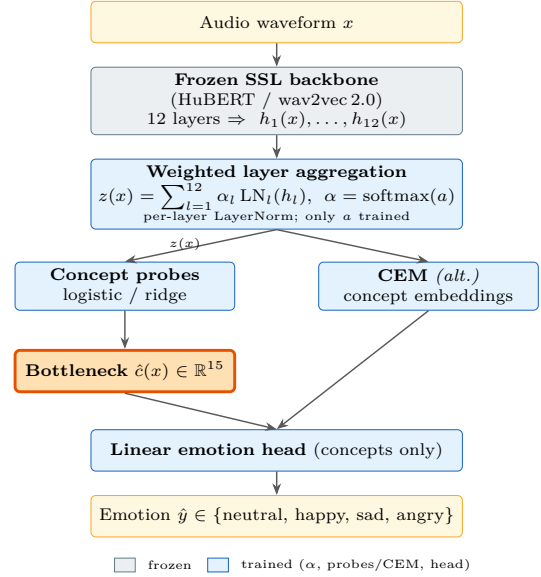


Figure 1: Pipeline overview. A frozen SSL backbone (HuBERT or wav2vec 2.0) produces 12 layer representations, normalized per layer by LayerNorm and combined by a trained weighted sum into an utterance embedding $z(x)$. Linear probes read z into 15 interpretable concepts (the bottleneck); a linear head predicts the emotion from the concepts only. The Concept Embedding Model is the alternative, vectorial concept channel. Shaded (grey) blocks are frozen; coloured (blue) blocks, the aggregation weights, probes/CEM and head, are the only trained parameters.

Table 1: Concept set. Expected correlation signs from the vocal emotion literature [3, 7, 14] (h = happy, s = sad, a = angry); n/a: not scored (mixed evidence).

Concept	Definition (unit)	Expect.
pitch_mean	mean F0, voiced frames (Hz)	h+ s- a+
pitch_std	F0 standard deviation (Hz)	n/a
pitch_range	F0 p90–p10 (semitones)	h+ s- a+
pitch_slope	F0 contour slope (st/s)	n/a
voiced_ratio	fraction of voiced frames	n/a
speech_rate	syllable-nuclei proxy rate (1/s)	s- a+
pause_ratio	internal silence fraction	s+
pause_dur_mean	mean internal pause (s)	n/a
rms_mean	mean RMS energy	h+ s- a+
rms_cv	RMS coeff. of variation	n/a
rms_slope	relative energy slope (1/s)	n/a
jitter	local F0 perturbation	n/a
shimmer	local amplitude perturbation	n/a
hnr	harmonics-to-noise ratio (dB)	n/a
alpha_ratio	50–1000 Hz vs. 1–5 kHz energy (dB)	s+ a-

on the full training set. Concept ground truth is binarized by per-gender median thresholds estimated on training data only: this removes the gender confound on pitch while remaining deployment-valid (gender is annotated in IEMOCAP and would be inferred at deployment). We evaluate three bottleneck variants: *soft* (probe probabilities), *hard* (thresholded 0/1 predictions), and *continuous*

(ridge-regression probes on per-gender z-scored concept values). The emotion head is a class-balanced logistic regression on the concept vector. Probe accuracy averages 0.78 on both backbones, but per-concept regression quality varies widely (r^2 from 0.89 for rms_cv down to 0.10 for pitch_slope): a concept the backbone barely encodes contributes little and should be read as a weak explanatory unit rather than a reliable one. Soft and continuous channels can carry label information beyond the concepts’ semantics (the *leakage* problem [10]), which we anticipate here and quantify in Section 5.

4.5 Concept Interventions and Completeness

Following [9], interventions progressively replace predicted concepts with ground truth, in random order or by probe uncertainty (least confident first). We add a *targeted* policy that corrects only the concepts the probes actually got wrong; we introduce it as a localization instrument for leakage rather than as a performance booster. Two oracles retrain the head directly on ground-truth concepts, binary (OB) and continuous (OC), giving the decomposition: incompleteness = $\text{Acc}_{\text{BB}} - \text{Acc}_{\text{OC}}$ and binarization cost = $\text{Acc}_{\text{OC}} - \text{Acc}_{\text{OB}}$, where BB is the strongest unconstrained model on the same representation, i.e., the linear baseline.

4.6 Concept Embedding Model

The CEM [17] is trained on the same aggregated embedding: each concept k has two 16-dimensional embeddings c_k^+, c_k^- mixed by a supervised scalar score p_k (BCE on the pre-intervention logits), and a joint linear head reads the concatenated mixture. Training uses RandInt [17]: each concept of each sample is intervened ($p_k :=$ ground truth) with probability 0.25, so that the head learns to exploit corrections; no intervention is applied at evaluation. The CEM is entirely by-design (the frozen embedding is merely its input); its channel is vectorial, supervision and interventions remain scalar.

5 EXPERIMENTS AND ANALYSIS

5.1 Dataset and Experimental Setup

IEMOCAP [4]: 5,531 utterances, four classes (excited merged into happy; other labels discarded for comparability with the standard 4-class setup). The split is speaker-independent: Session 5 (two unseen speakers) is the held-out test set, never touched before final evaluation; Sessions 1–4 form the training pool, within which Session 4 is the internal validation that drives *all* model selection (layer choice, regularization strengths, and early stopping with weight restore). Early-stopped models, the layer aggregator, the MLP, and the CEM, are fit on Sessions 1–3 and validated on Session 4; the logistic and ridge probes and the emotion heads select their regularization on Session 4 and are then refit on the full pool (Sessions 1–4). We report weighted accuracy (WA) and unweighted accuracy (UA, macro recall, standard in SER given class imbalance); macro-F1 tracks UA closely and is omitted for space. Significance uses exact McNemar tests and bootstrap confidence intervals on paired predictions.

5.2 Baselines and Ablations

On the same aggregated embedding, the linear baseline reaches 0.670/0.658 accuracy (HuBERT/wav2vec2), an MLP 0.656/0.653, and

Table 2: Emotion classification on the test session (Session 5), ordered by accuracy. Acc (UA) for HuBERT and wav2vec2. Oracles are retrained on ground-truth concepts and depend only on them, hence identical across backbones.

Model	HuBERT	wav2vec2
Oracle (binary GT concepts)	0.421 (0.479)	0.421 (0.479)
CBM hard	0.438 (0.507)	0.435 (0.503)
CBM soft	0.467 (0.526)	0.459 (0.518)
Oracle (continuous GT)	0.470 (0.501)	0.470 (0.501)
CBM continuous	0.484 (0.532)	0.504 (0.537)
CBM + V/A/D (diagnostic)	0.602 (0.620)	0.607 (0.621)
MLP	0.656 (0.664)	0.653 (0.651)
CEM	0.668 (0.676)	0.643 (0.642)
Linear baseline	0.670 (0.677)	0.658 (0.670)

an MLP on the 12 concatenated layers 0.645/0.647, consistent with SUPERB-level probing. An attention-pooling classifier over 8 temporal chunks reaches 0.579/0.604 and provides a temporal breakdown of salient segments. Ablations: adding the four MFCC concepts to the soft CBM buys at most one point (from 0.467 to 0.476 on HuBERT, from 0.459 to 0.462 on wav2vec2): the uninterpretable concepts add nothing usable; adding valence/activation/dominance jumps to 0.602/0.607, confirming they are quasi-labels that belong outside the model. The concept set was extended from K=9 after measuring its ceiling (binary oracle 0.391→0.421)

5.3 Main Results

Table 2 contains the central result; the CBM-vs-MLP gap is real, not noise (McNemar $p < 10^{-26}$ on both backbones). Three observations organize the discussion.

The hierarchy is a spectrum between explainability and accuracy. All rows share the same representation; they differ only in how constrained the concept channel is. From hard CBM through soft and continuous scores to the CEM (vector channel), every relaxation buys accuracy and pays in explainability: the CEM is statistically indistinguishable from the MLP on both backbones (McNemar $p=0.15/0.18$) and ≈ 20 points above the soft CBM ($p < 10^{-24}$), but its channel can no longer be read as “these concepts, and nothing else”.

Incompleteness dominates the cost. The double-oracle decomposition gives ≈ 5 points for binarization (0.470–0.421) and ≈ 20 points for incompleteness (linear baseline 0.670 vs. continuous oracle 0.470), a 4:1 ratio. The vocabulary, not the discretization, is the bottleneck: extending it from 9 to 15 concepts moved the oracle by only ≈ 3 points, consistent with the modest accuracy typically reported for fixed acoustic parameter sets such as eGeMAPS [5]. What is missing is what utterance-level functionals cannot express: temporal dynamics and lexical content, which the SSL embedding encodes. The diagnostic V/A/D row (0.602) confirms this from the other side: give the head a quasi-label vocabulary and the gap closes.

Soft leakage is real and localized. The soft CBM *outperforms* the binary oracle (0.467 vs. 0.421), which is impossible unless the soft scores carry extra-concept information. The hard CBM, instead, is statistically indistinguishable from the oracle on both backbones (McNemar $p=0.29/0.40$; bootstrap 95% CI of the difference includes 0): binarization reduces the leak to statistical insignificance, though

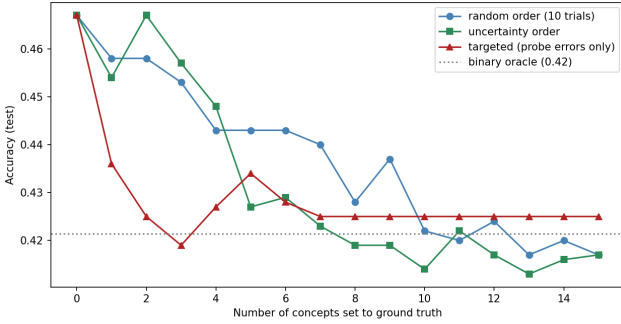


Figure 2: Test accuracy while replacing n predicted concepts with ground truth (HuBERT): random, uncertainty and targeted order (only probe errors), with the binary-oracle level. The targeted curve reaches the oracle at $n \approx 3$.

the hard CBM still edges the oracle by +1.7/+1.4 points. The continuous channel leaks as well, CBM continuous exceeds its own (continuous) oracle by +1.5 points on HuBERT and +3.5 on wav2vec2, but the largest excess is the soft channel over its binary oracle (+4.5/+3.7): every relaxation of the concept channel, soft or continuous, lets task-correlated information through. Section 5.4 shows where the leak lives.

5.4 Concept Intervention Analysis

All CBM intervention curves *descend* (Figure 2; random order: from 0.467 to 0.417 on HuBERT, from 0.459 to 0.425 on wav2vec2). This is a completeness measurement, not a controllability failure: since the soft scores already exceed the oracle, replacing them with ground truth strips the leaked information instead of adding knowledge. The descent speed (targeted > uncertainty > random) tracks how much leaked information each policy touches. The targeted curve converges to the oracle at $n \approx 3$ corrected concepts, matching the mean of ≈ 3.4 probe errors per sample. Endpoints dip slightly below the oracle because the injected bits cross a scaler fitted on soft scores, unlike the retrained oracle. On the subgroup with at least one probe error, accuracy drops from 0.460 to 0.418 (HuBERT) and to 0.435 (wav2vec2): causal evidence that the leakage lives precisely in the probes’ errors, a phenomenon we call *leakage localization*. Effective interventions need mispredicted concepts and a complete vocabulary; we only have the former. The CEM accepts corrections without degrading but has no headroom: random, uncertainty and targeted curves are all flat (from 0.668 to 0.659 and from 0.643 to 0.640; targeted to 0.663/0.641); its concept scores are about as accurate as the probes (mean 0.757/0.728), so flatness reflects a model already at black-box level, not corrections being ignored.

5.5 Explanation Quality (Co-12)

We group our measurements under the Co-12 properties [12]; our *faithfulness* corresponds to Co-12 Correctness and *plausibility* to Coherence. Values below are reported as HuBERT/wav2vec2 on the soft CBM.

Faithfulness. AOPC comprehensiveness 0.228/0.233 vs. random-order 0.149/0.157; sufficiency 0.462/0.462 vs. 0.381/0.388. For a linear

head over concepts these deletion metrics are structural (faithful by design); we read them as sanity checks. A surrogate distilled on the black-box’s predictions matches it on 0.587/0.563 of test samples: the concept vocabulary approximates the black-box only partially (incompleteness again). Randomization checks [1] pass: importance rankings correlate at 0.046/0.093 with label-randomized models (empirical null $\approx 0.19/0.24$) and at $-0.05/-0.07$ under parameter randomization.

Consistency. Full leave-one-session-out retraining: accuracy 0.462 ± 0.020 and 0.463 ± 0.019 , importance rank-correlation 0.88/0.87 across folds (same top concept in every fold).

Contrastivity. Null, and we report it as such: mean off-diagonal cosine between class weight vectors $-0.279/-0.273$, no better than label-shuffled models ($-0.308/-0.319$; structural baseline $-1/3$). Per-class signatures do not separate beyond what the softmax geometry imposes.

Covariate complexity. Mean absolute off-diagonal correlation between concept scores 0.201/0.200: contained redundancy.

Controllability. Section 5.4.

Plausibility. 11 of 14 literature expectations (Table 1) are matched on both backbones; e.g. `alpha_ratio` behaves as predicted [3] (angry -0.29 , sad $+0.27/+0.30$). The main violation is instructive: sad correlates *negatively* with `pause_ratio` on the predicted concept scores ($-0.21/-0.23$) but near zero on the raw features (-0.04): the probes reconstruct the concept using task-correlated information, i.e., the leakage of Section 5.4 seen from another angle.

5.6 Backbone Comparison: HuBERT vs. wav2vec 2.0

Running the identical pipeline on wav2vec 2.0 replicates every key finding: the hierarchy keeps its shape (Table 2), hard $\approx$ oracle ($p = 0.40$), leakage localizes in probe errors, the CEM is intervention-flat, contrastivity is null, the sad/pause paradox persists, plausibility is 11/14, and the probe- r^2 profile is superimposable (rms_cv and hnr best, pitch_slope worst on both). The oracles are identical across backbones, as they must be since they depend only on ground-truth concepts, certifying the pipeline end-to-end. Minor differences, best layer 9 vs. 7 when selected on validation UA (on wav2vec2 accuracy would pick layer 9: a live example of why UA drives selection), a slightly weaker CEM, a leakier continuous channel on wav2vec2 and a stronger temporal attention on wav2vec2, do not affect any qualitative conclusion: the observed phenomena are properties of the concept-based setup rather than of a single representation.

6 CONCLUSIONS

We measured, rather than asserted, the cost of interpretability in SER on frozen SSL representations. The bottleneck variants form a spectrum between explainability and accuracy; the cost is dominated by concept-set incompleteness rather than binarization; the leakage of unconstrained channels is causally localized in the probes’ errors; and the CEM is the practical midpoint: black-box accuracy, weaker explainability. Limitations: V/A/D are partially definitional of the target, hence excluded; IEMOCAP is acted speech; the two backbones are architecturally similar; global acoustic functionals saturate, so future work targets temporal and lexical concepts.

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